

Semantics of Comparative Constructions

Class 5: Quantifiers in *than*-clauses II

Filipe Hisao Kobayashi

University of Salzburg

3 August 2026

OVA Summer School · University of Debrecen

Part 1

Comparing our comparative operators

Comparing two meanings

- In this class, we've assigned two different truth conditions to simple comparative clauses:

(1) Ann is taller than Beth.

Comparing two meanings

- In this class, we've assigned two different truth conditions to simple comparative clauses:

(1) Ann is taller than Beth.

a. **Proper containment**

TALL(beth) $\subset$ TALL(ann)

Comparing two meanings

- In this class, we've assigned two different truth conditions to simple comparative clauses:

(1) Ann is taller than Beth.

a. **Proper containment**

$$\mathbf{TALL(beth)} \subset \mathbf{TALL(ann)}$$

b. **Difference**

$$\mathbf{TALL(ann)} - \mathbf{TALL(beth)} \neq \emptyset$$

Comparing two meanings

- In this class, we've assigned two different truth conditions to simple comparative clauses:

(1) Ann is taller than Beth.

a. **Proper containment**

$$\mathbf{TALL(beth)} \subset \mathbf{TALL(ann)}$$

b. **Difference**

$$\mathbf{TALL(ann)} - \mathbf{TALL(beth)} \neq \emptyset$$

$$\Leftrightarrow \exists d [d \in \mathbf{TALL(ann)} \wedge d \notin \mathbf{TALL(beth)}]$$

$$\mathbf{TALL}(x) := \{d \mid \mathbf{height}(x) \geq d\}$$

Comparing two meanings

- In this class, we've assigned two different truth conditions to simple comparative clauses:

(1) Ann is taller than Beth.

a. **Proper containment**

$$\mathbf{TALL(beth)} \subset \mathbf{TALL(ann)}$$

b. **Difference**

$$\mathbf{TALL(ann)} - \mathbf{TALL(beth)} \neq \emptyset$$

$$\Leftrightarrow \exists d [d \in \mathbf{TALL(ann)} \wedge d \notin \mathbf{TALL(beth)}]$$

$$\mathbf{TALL}(x) := \{d \mid \mathbf{height}(x) \geq d\}$$

- These are **not equivalent** in general.

Comparing two meanings

- In this class, we've assigned two different truth conditions to simple comparative clauses:

(1) Ann is taller than Beth.

a. **Proper containment**

$$\mathbf{TALL(beth)} \subset \mathbf{TALL(ann)}$$

b. **Difference**

$$\mathbf{TALL(ann)} - \mathbf{TALL(beth)} \neq \emptyset$$

$$\Leftrightarrow \exists d [d \in \mathbf{TALL(ann)} \wedge d \notin \mathbf{TALL(beth)}]$$

$$\mathbf{TALL}(x) := \{d \mid \mathbf{height}(x) \geq d\}$$

- These are **not equivalent** in general.
- Proper containment requires **two** things:

(2)

$$\begin{aligned} \mathbf{TALL(beth)} \subset \mathbf{TALL(ann)} \quad \Leftrightarrow \quad & \underbrace{\forall d [d \in \mathbf{TALL(beth)} \rightarrow d \in \mathbf{TALL(ann)}]}_{\text{containment}} \\ & \wedge \underbrace{\exists d [d \in \mathbf{TALL(ann)} \wedge d \notin \mathbf{TALL(beth)}]}_{\text{non-empty difference}} \end{aligned}$$

Does it matter though?

- The crucial case would be one where:

1) $D_{\text{matrix}} - D_{\text{than}} \neq \emptyset$

2) $D_{\text{than}} \not\subseteq D_{\text{matrix}}$

Does it matter though?

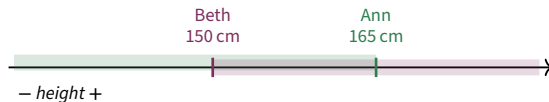
- The crucial case would be one where:

1) $D_{\text{matrix}} - D_{\text{than}} \neq \emptyset$

2) $D_{\text{than}} \not\subseteq D_{\text{matrix}}$

- For example:

(3) Ann is taller than Beth is short.



Does it matter though?

- The crucial case would be one where:

1) $D_{\text{matrix}} - D_{\text{than}} \neq \emptyset$

2) $D_{\text{than}} \not\subseteq D_{\text{matrix}}$

- For example:

(3) Ann is taller than Beth is short.

non-empty difference



- The difference is non-empty, but neither interval contains the other.

Does it matter though?

- The crucial case would be one where:

1) $D_{\text{matrix}} - D_{\text{than}} \neq \emptyset$

2) $D_{\text{than}} \not\subseteq D_{\text{matrix}}$

- For example:

(3) Ann is taller than Beth is short.

non-empty difference



- The difference is non-empty, but neither interval contains the other.
- But our **commensurability condition** already rules out precisely such cases:

(4) $\llbracket \text{er} \rrbracket(D)(D')$ is defined only if $D \subseteq D'$ or $D' \subseteq D$.

Does it matter though?

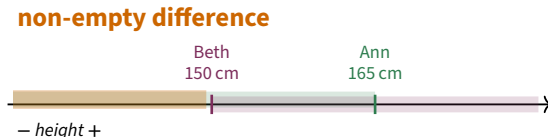
- The crucial case would be one where:

1) $D_{\text{matrix}} - D_{\text{than}} \neq \emptyset$

2) $D_{\text{than}} \not\subseteq D_{\text{matrix}}$

- For example:

(3) Ann is taller than Beth is short.



- The difference is non-empty, but neither interval contains the other.
- But our **commensurability condition** already rules out precisely such cases:

(4) $\llbracket \text{er} \rrbracket(D)(D')$ is defined only if $D \subseteq D'$ or $D' \subseteq D$.

⇒ Whenever *er* is defined, the two analyses are equivalent.

Part 2

The problem of quantifiers in *than*-clauses

Part 2

The problem of quantifiers in *than*-clauses

What works: disjunction (recap)

What doesn't work: conjunction

Can QR fix this?

Disjunction I

- As we discussed last class, we can account for disjunction in *than*-clauses:

(5) Ann is older than either Beth or Cleo are.

$\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

Disjunction I

- As we discussed last class, we can account for disjunction in *than*-clauses:

(5) Ann is older than either Beth or Cleo are.

$\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(6) a. $\text{than } \lambda_d \text{ } [[\text{Beth is } d\text{-old}] \text{ or } [\text{Cleo is } d\text{-old}]]$

b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \vee (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

Disjunction I

- As we discussed last class, we can account for disjunction in *than*-clauses:

(5) Ann is older than either Beth or Cleo are.

$\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(6) a. $\text{than } \lambda_d \text{ } [[\text{Beth is } d\text{-old}] \text{ or } [\text{Cleo is } d\text{-old}]]$

b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \vee (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

- Examples:

- Beth is 2 years old, Cleo is 3 years old:

$D_{\text{than}} = \{ \quad \quad \quad \}$

Disjunction I

- As we discussed last class, we can account for disjunction in *than*-clauses:

(5) Ann is older than either Beth or Cleo are.

$\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(6) a. $\text{than } \lambda_d [[\text{Beth is } d\text{-old}] \text{ or } [\text{Cleo is } d\text{-old}]]$

b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \vee (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

- Examples:

- Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}} \quad \quad \quad \}$$

Disjunction I

- As we discussed last class, we can account for disjunction in *than*-clauses:

(5) Ann is older than either Beth or Cleo are.

$\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(6) a. $\text{than } \lambda_d [[\text{Beth is } d\text{-old}] \text{ or } [\text{Cleo is } d\text{-old}]]$

b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \vee (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

- Examples:

- Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}} \}$$

Disjunction I

- As we discussed last class, we can account for disjunction in *than*-clauses:

(5) Ann is older than either Beth or Cleo are.

$\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(6) a. $\text{than } \lambda_d [[\text{Beth is } d\text{-old}] \text{ or } [\text{Cleo is } d\text{-old}]]$

b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \vee (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

- Examples:

- Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}}, d_{\text{three}}\}$$

Disjunction I

- As we discussed last class, we can account for disjunction in *than*-clauses:

(5) Ann is older than either Beth or Cleo are.
 $\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(6) a. $\text{than } \lambda_d \text{ [[Beth is } d\text{-old] or [Cleo is } d\text{-old]]}$
b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \vee (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

- Examples:

- Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}}, d_{\text{three}}\}$$

- Beth is 4 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{ \quad \quad \quad \}$$

Disjunction I

- As we discussed last class, we can account for disjunction in *than*-clauses:

(5) Ann is older than either Beth or Cleo are.
 $\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(6) a. $\text{than } \lambda_d [[\text{Beth is } d\text{-old}] \text{ or } [\text{Cleo is } d\text{-old}]]$
b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \vee (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

- Examples:

- Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}}, d_{\text{three}}\}$$

- Beth is 4 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}} \quad \quad \quad \}$$

Disjunction I

- As we discussed last class, we can account for disjunction in *than*-clauses:

(5) Ann is older than either Beth or Cleo are.

$\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(6) a. $\text{than } \lambda_d [[\text{Beth is } d\text{-old}] \text{ or } [\text{Cleo is } d\text{-old}]]$

b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \vee (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

- Examples:

- Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}}, d_{\text{three}}\}$$

- Beth is 4 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}} \quad \quad \quad \}$$

Disjunction I

- As we discussed last class, we can account for disjunction in *than*-clauses:

(5) Ann is older than either Beth or Cleo are.

$\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(6) a. $\text{than } \lambda_d \text{ [[Beth is } d\text{-old] or [Cleo is } d\text{-old]]}$

b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \vee (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

- Examples:

- Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}}, d_{\text{three}}\}$$

- Beth is 4 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}}, d_{\text{three}} \quad \}$$

Disjunction I

- As we discussed last class, we can account for disjunction in *than*-clauses:

(5) Ann is older than either Beth or Cleo are.

$\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(6) a. $\text{than } \lambda_d \text{ [[Beth is } d\text{-old] or [Cleo is } d\text{-old]]}$

b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \vee (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

- Examples:

- Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}}, d_{\text{three}}\}$$

- Beth is 4 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}}, d_{\text{three}}, d_{\text{four}}\}$$

Disjunction I

- As we discussed last class, we can account for disjunction in *than*-clauses:

(5) Ann is older than either Beth or Cleo are.
 $\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(6) a. $\text{than } \lambda_d [[\text{Beth is } d\text{-old}] \text{ or } [\text{Cleo is } d\text{-old}]]$
b. $D_{\text{than}} := \{d \mid (\text{age}(\text{beth}) \geq d) \vee (\text{age}(\text{cleo}) \geq d)\}$

- Examples:

- Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}}, d_{\text{three}}\} = \text{AGE}(\text{cleo})$$

- Beth is 4 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}}, d_{\text{three}}, d_{\text{four}}\} = \text{AGE}(\text{beth})$$

$\Rightarrow$ the *than*-clause will denote the interval of whoever's oldest!

Disjunction II

- This follows from some elementary facts about set theory:

$$\begin{aligned}(7) \quad & \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \vee (\mathbf{age}(\mathbf{cleo}) \geq d)\} \\ &= \{d \mid \mathbf{age}(\mathbf{beth}) \geq d\} \cup \{d \mid \mathbf{age}(\mathbf{cleo}) \geq d\}\end{aligned}$$

Disjunction II

- This follows from some elementary facts about set theory:

$$\begin{aligned}(7) \quad & \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \vee (\mathbf{age}(\mathbf{cleo}) \geq d)\} \\ &= \{d \mid \mathbf{age}(\mathbf{beth}) \geq d\} \cup \{d \mid \mathbf{age}(\mathbf{cleo}) \geq d\}\end{aligned}$$

$$(8) \quad \text{If } A \subseteq B, \text{ then } A \cup B = B$$

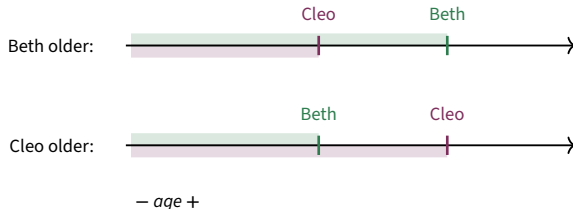
Disjunction II

- This follows from some elementary facts about set theory:

$$(7) \quad \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \vee (\mathbf{age}(\mathbf{cleo}) \geq d)\} \\ = \{d \mid \mathbf{age}(\mathbf{beth}) \geq d\} \cup \{d \mid \mathbf{age}(\mathbf{cleo}) \geq d\}$$

$$(8) \quad \text{If } A \subseteq B, \text{ then } A \cup B = B$$

- Illustration:



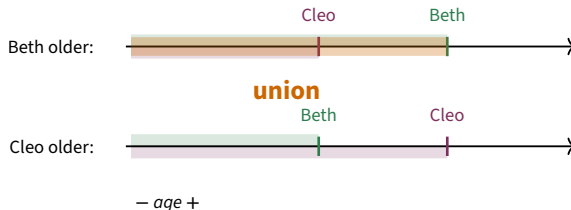
Disjunction II

- This follows from some elementary facts about set theory:

$$(7) \quad \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \vee (\mathbf{age}(\mathbf{cleo}) \geq d)\} \\ = \{d \mid \mathbf{age}(\mathbf{beth}) \geq d\} \cup \{d \mid \mathbf{age}(\mathbf{cleo}) \geq d\}$$

$$(8) \quad \text{If } A \subseteq B, \text{ then } A \cup B = B$$

- Illustration:



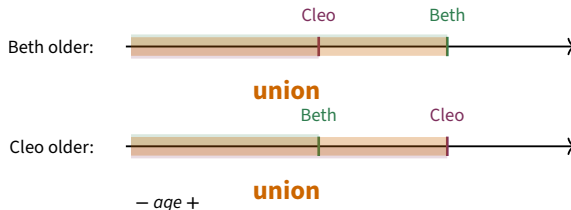
Disjunction II

- This follows from some elementary facts about set theory:

$$(7) \quad \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \vee (\mathbf{age}(\mathbf{cleo}) \geq d)\} \\ = \{d \mid \mathbf{age}(\mathbf{beth}) \geq d\} \cup \{d \mid \mathbf{age}(\mathbf{cleo}) \geq d\}$$

$$(8) \quad \text{If } A \subseteq B, \text{ then } A \cup B = B$$

- Illustration:



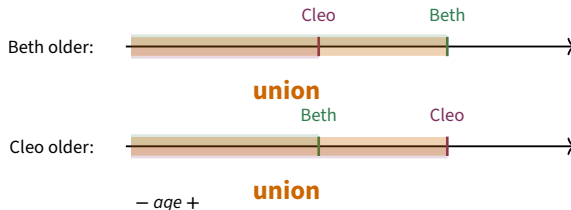
Disjunction II

- This follows from some elementary facts about set theory:

$$(7) \quad \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \vee (\mathbf{age}(\mathbf{cleo}) \geq d)\} \\ = \{d \mid \mathbf{age}(\mathbf{beth}) \geq d\} \cup \{d \mid \mathbf{age}(\mathbf{cleo}) \geq d\}$$

$$(8) \quad \text{If } A \subseteq B, \text{ then } A \cup B = B$$

- Illustration:



⇒ No matter who is older, *than Beth or Cleo* denotes the interval of the **older** individual.

Disjunction III

- In the general case, our analysis seems to work:

(9) **What we want**

$\llbracket \text{A is older than either B or C.} \rrbracket \approx \text{A is older than B \& A is older than C}$

(10) **What we get**

$\llbracket \text{A is older than either B or C.} \rrbracket \approx \text{A is older than whoever among B and C is older.}$

Disjunction III

- In the general case, our analysis seems to work:

(9) **What we want**

$\llbracket A \text{ is older than either } B \text{ or } C. \rrbracket \approx A \text{ is older than } B \ \& \ A \text{ is older than } C$

(10) **What we get**

$\llbracket A \text{ is older than either } B \text{ or } C. \rrbracket \approx A \text{ is older than whoever among } B \text{ and } C \text{ is older.}$

- When they come apart is on that datapoint we discussed last time:

(11) **What we want**

$\llbracket A \text{ is at most 2 years older than either } B \text{ or } C. \rrbracket$

$\approx A \text{ is at most 2 years older than } B \ \& \ A \text{ is at most 2 years older than } C$

(12) **What we get**

$\llbracket A \text{ is at most 2 years older than either } B \text{ or } C. \rrbracket$

$\approx A \text{ is at most 2 years older than whoever among } B \text{ and } C \text{ is older}$

Part 2

The problem of quantifiers in *than*-clauses

What works: disjunction (recap)

What doesn't work: conjunction

Can QR fix this?

Conjunction I

- A sentence we most definitely should be able to account for:

(13) Ann is older than Beth and Cleo (both) are.

$\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

Conjunction I

- A sentence we most definitely should be able to account for:

(13) Ann is older than Beth and Cleo (both) are.
 $\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(14) a. $\text{than } \lambda_d \text{ [[Beth is } d\text{-old] and [Cleo is } d\text{-old]]}$
b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \wedge (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

Conjunction I

- A sentence we most definitely should be able to account for:

(13) Ann is older than Beth and Cleo (both) are.
 $\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(14) a. $\text{than } \lambda_d [[\text{Beth is } d\text{-old}] \text{ and } [\text{Cleo is } d\text{-old}]]$
b. $D_{\text{than}} := \{d \mid (\text{age}(\text{beth}) \geq d) \wedge (\text{age}(\text{cleo}) \geq d)\}$

- Examples:
 - Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{ \quad \quad \quad \}$$

Conjunction I

- A sentence we most definitely should be able to account for:

(13) Ann is older than Beth and Cleo (both) are.
 $\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(14) a. $\text{than } \lambda_d [[\text{Beth is } d\text{-old}] \text{ and } [\text{Cleo is } d\text{-old}]]$
b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \wedge (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

- Examples:
 - Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}} \quad \quad \quad \}$$

Conjunction I

- A sentence we most definitely should be able to account for:

(13) Ann is older than Beth and Cleo (both) are.
 $\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(14) a. $\text{than } \lambda_d [[\text{Beth is } d\text{-old}] \text{ and } [\text{Cleo is } d\text{-old}]]$
b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \wedge (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

- Examples:
 - Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}} \quad \}$$

Conjunction I

- A sentence we most definitely should be able to account for:

(13) Ann is older than Beth and Cleo (both) are.
 $\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(14) a. $\text{than } \lambda_d [[\text{Beth is } d\text{-old}] \text{ and } [\text{Cleo is } d\text{-old}]]$
b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \wedge (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

- Examples:
 - Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}} \quad \}$$

Conjunction I

- A sentence we most definitely should be able to account for:

(13) Ann is older than Beth and Cleo (both) are.
 $\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(14) a. $\text{than } \lambda_d [[\text{Beth is } d\text{-old}] \text{ and } [\text{Cleo is } d\text{-old}]]$
b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \wedge (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

- Examples:
 - Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}} \}$$

- Beth is 4 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{ \}$$

Conjunction I

- A sentence we most definitely should be able to account for:

(13) Ann is older than Beth and Cleo (both) are.
 $\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(14) a. $\text{than } \lambda_d \text{ [[Beth is } d\text{-old] and [Cleo is } d\text{-old]]}$
b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \wedge (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

- Examples:
 - Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}} \}$$

- Beth is 4 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}} \}$$

Conjunction I

- A sentence we most definitely should be able to account for:

(13) Ann is older than Beth and Cleo (both) are.
 $\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(14) a. $\text{than } \lambda_d [[\text{Beth is } d\text{-old}] \text{ and } [\text{Cleo is } d\text{-old}]]$
b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \wedge (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

- Examples:
 - Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}} \}$$

- Beth is 4 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}} \}$$

Conjunction I

- A sentence we most definitely should be able to account for:

(13) Ann is older than Beth and Cleo (both) are.
 $\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(14) a. $\text{than } \lambda_d [[\text{Beth is } d\text{-old}] \text{ and } [\text{Cleo is } d\text{-old}]]$
b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \wedge (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

- Examples:
 - Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}} \quad \}$$

- Beth is 4 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}}, d_{\text{three}} \quad \}$$

Conjunction I

- A sentence we most definitely should be able to account for:

(13) Ann is older than Beth and Cleo (both) are.
 $\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(14) a. $\text{than } \lambda_d \text{ [[Beth is } d\text{-old] and [Cleo is } d\text{-old]]}$
b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \wedge (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

- Examples:
 - Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}} \quad \}$$

- Beth is 4 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}}, d_{\text{three}} \quad \}$$

Conjunction I

- A sentence we most definitely should be able to account for:

(13) Ann is older than Beth and Cleo (both) are.
 $\rightsquigarrow$ *Ann is taller than Beth and Ann is taller than Cleo*

- The *than*-clause is interpreted as follows:

(14) a. $\text{than } \lambda_d [[\text{Beth is } d\text{-old}] \text{ and } [\text{Cleo is } d\text{-old}]]$
b. $D_{\text{than}} := \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \wedge (\mathbf{age}(\mathbf{cleo}) \geq d)\}$

- Examples:
 - Beth is 2 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}} \}$$

- Beth is 4 years old, Cleo is 3 years old:

$$D_{\text{than}} = \{d_{\text{one}}, d_{\text{two}}, d_{\text{three}} \}$$

$\Rightarrow$ the *than*-clause will denote the interval of whoever's youngest!

Conjunction II

- This follows from some elementary facts about set theory:

$$\begin{aligned}(15) \quad & \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \wedge (\mathbf{age}(\mathbf{cleo}) \geq d)\} \\ &= \{d \mid \mathbf{age}(\mathbf{beth}) \geq d\} \cap \{d \mid \mathbf{age}(\mathbf{cleo}) \geq d\}\end{aligned}$$

Conjunction II

- This follows from some elementary facts about set theory:

$$\begin{aligned}(15) \quad & \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \wedge (\mathbf{age}(\mathbf{cleo}) \geq d)\} \\ &= \{d \mid \mathbf{age}(\mathbf{beth}) \geq d\} \cap \{d \mid \mathbf{age}(\mathbf{cleo}) \geq d\}\end{aligned}$$

$$(16) \quad \text{If } A \subseteq B, \text{ then } A \cap B = A.$$

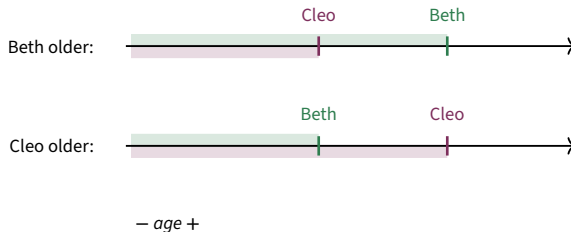
Conjunction II

- This follows from some elementary facts about set theory:

$$(15) \quad \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \wedge (\mathbf{age}(\mathbf{cleo}) \geq d)\} \\ = \{d \mid \mathbf{age}(\mathbf{beth}) \geq d\} \cap \{d \mid \mathbf{age}(\mathbf{cleo}) \geq d\}$$

$$(16) \quad \text{If } A \subseteq B, \text{ then } A \cap B = A.$$

- Illustration:



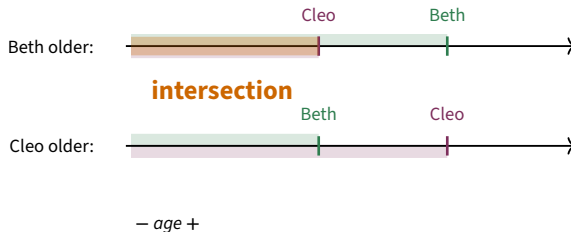
Conjunction II

- This follows from some elementary facts about set theory:

$$(15) \quad \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \wedge (\mathbf{age}(\mathbf{cleo}) \geq d)\} \\ = \{d \mid \mathbf{age}(\mathbf{beth}) \geq d\} \cap \{d \mid \mathbf{age}(\mathbf{cleo}) \geq d\}$$

$$(16) \quad \text{If } A \subseteq B, \text{ then } A \cap B = A.$$

- Illustration:



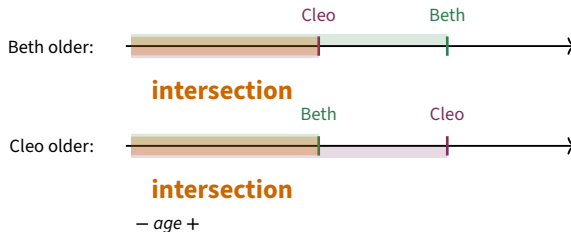
Conjunction II

- This follows from some elementary facts about set theory:

$$(15) \quad \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \wedge (\mathbf{age}(\mathbf{cleo}) \geq d)\} \\ = \{d \mid \mathbf{age}(\mathbf{beth}) \geq d\} \cap \{d \mid \mathbf{age}(\mathbf{cleo}) \geq d\}$$

$$(16) \quad \text{If } A \subseteq B, \text{ then } A \cap B = A.$$

- Illustration:



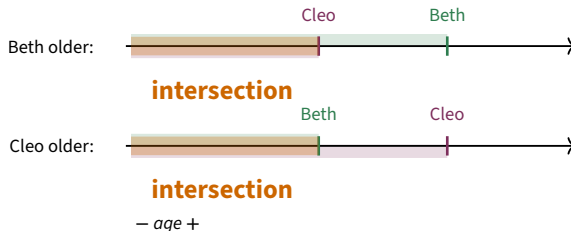
Conjunction II

- This follows from some elementary facts about set theory:

$$(15) \quad \{d \mid (\mathbf{age}(\mathbf{beth}) \geq d) \wedge (\mathbf{age}(\mathbf{cleo}) \geq d)\} \\ = \{d \mid \mathbf{age}(\mathbf{beth}) \geq d\} \cap \{d \mid \mathbf{age}(\mathbf{cleo}) \geq d\}$$

$$(16) \quad \text{If } A \subseteq B, \text{ then } A \cap B = A.$$

- Illustration:



$\Rightarrow$ *than Beth and Cleo are* denotes the interval of the **younger** individual.

Conjunction III

- For conjunction, we fail miserably:

(17) **What we want**

$\llbracket A \text{ is older than either } B \text{ and } C. \rrbracket \approx A \text{ is older than } B \ \& \ A \text{ is older than } C$

(18) **What we get**

$\llbracket A \text{ is older than either } B \text{ or } C. \rrbracket \approx A \text{ is older than whoever among } B \text{ and } C \text{ is younger.}$

Every

- For the same reason, we make wrong predictions regarding every:

(19) Ann is older than every boy.

$\rightsquigarrow$ *for every boy x : Ann is older than x*

Every

- For the same reason, we make wrong predictions regarding every:

(19) Ann is older than every boy.

$\rightsquigarrow$ *for every boy x : Ann is older than x*

- The denotation of the *than*-clause:

(20) a. $\text{than } \lambda_d \text{ every boy is } d\text{-old}$

b. $D_{\text{than}} := \{d \mid \forall x[\mathbf{boy}(x) \rightarrow \mathbf{age}(x) \geq d]\}$

Every

- For the same reason, we make wrong predictions regarding every:

(19) Ann is older than every boy.

$\rightsquigarrow$ for every boy x : Ann is older than x

- The denotation of the *than*-clause:

(20) a. $\text{than } \lambda_d \text{ every boy is } d\text{-old}$

b. $D_{\text{than}} := \{d \mid \forall x[\mathbf{boy}(x) \rightarrow \mathbf{age}(x) \geq d]\}$



Every

- For the same reason, we make wrong predictions regarding every:

(19) Ann is older than every boy.

$\rightsquigarrow$ for every boy x : Ann is older than x

- The denotation of the *than*-clause:

(20) a. $\text{than } \lambda_d \text{ every boy is } d\text{-old}$

b. $D_{\text{than}} := \{d \mid \forall x[\mathbf{boy}(x) \rightarrow \mathbf{age}(x) \geq d]\}$



Every

- For the same reason, we make wrong predictions regarding every:

(19) Ann is older than every boy.

$\rightsquigarrow$ for every boy x : Ann is older than x

- The denotation of the *than*-clause:

(20) a. $\text{than } \lambda_d \text{ every boy is } d\text{-old}$

b. $D_{\text{than}} := \{d \mid \forall x[\mathbf{boy}(x) \rightarrow \mathbf{age}(x) \geq d]\}$



Every

- For the same reason, we make wrong predictions regarding every:

(19) Ann is older than every boy.

$\rightsquigarrow$ for every boy x : Ann is older than x

- The denotation of the *than*-clause:

(20) a. $\text{than } \lambda_d \text{ every boy is } d\text{-old}$

b. $D_{\text{than}} := \{d \mid \forall x[\mathbf{boy}(x) \rightarrow \mathbf{age}(x) \geq d]\}$



$\Rightarrow$ the analysis predicts that every boy picks out the interval of the **youngest** boy.

- Since Ann is older than Bob, (19) is incorrectly predicted to be true.

Part 2

The problem of quantifiers in *than*-clauses

What works: disjunction (recap)

What doesn't work: conjunction

Can QR fix this?

QR out of the *than*-clause

- Suppose we were to allow DPs to QR out of *than*-clauses:

QR out of the *than*-clause

- Suppose we were to allow DPs to QR out of *than*-clauses:

(21) a. [Beth and Cleo] λ_x [Ann is taller than t_x are]

$\rightsquigarrow$ *Ann is taller than Beth & Ann is taller than Cleo*

b. every boy λ_x [Ann is taller than t_x are]

$\rightsquigarrow$ *for every boy x : Ann is taller than x*

QR out of the *than*-clause

- Suppose we were to allow DPs to QR out of *than*-clauses:

(21) a. [Beth and Cleo] λ_x [Ann is taller than t_x are]

$\rightsquigarrow$ *Ann is taller than Beth & Ann is taller than Cleo*

b. every boy λ_x [Ann is taller than t_x are]

$\rightsquigarrow$ *for every boy x : Ann is taller than x*

- That could work for phrasal comparatives, but we know clausal comparatives are islands:

(22) a. * Which boys is Ann taller than t are?

b. * Ann is taller than no boy is.

QR out of the *than*-clause

- Suppose we were to allow DPs to QR out of *than*-clauses:

(21) a. [Beth and Cleo] λ_x [Ann is taller than t_x are]

$\rightsquigarrow$ *Ann is taller than Beth & Ann is taller than Cleo*

b. every boy λ_x [Ann is taller than t_x are]

$\rightsquigarrow$ *for every boy x : Ann is taller than x*

- That could work for phrasal comparatives, but we know clausal comparatives are islands:

(22) a. * Which boys is Ann taller than t are?

b. * Ann is taller than no boy is.

- Schwarzschild and Wilkinson (2002) have an even better argument against this.

Expect in *than*-clauses

(23) Ann is taller than I predicted.

Expect in than-clauses

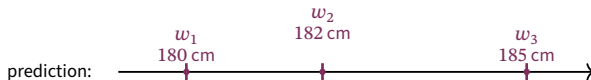
(23) Ann is taller than I predicted.

- **Context:** I predicted that Ann's height would be between 180 and 185 cm.

Expect in *than*-clauses

(23) Ann is taller than I predicted.

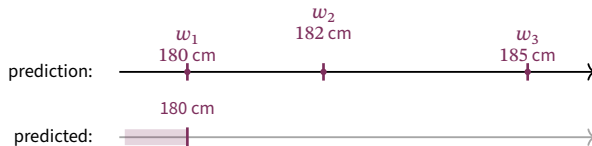
- **Context:** I predicted that Ann's height would be between 180 and 185 cm.
- What the sketched analysis predicts:



Expect in *than*-clauses

(23) Ann is taller than I predicted.

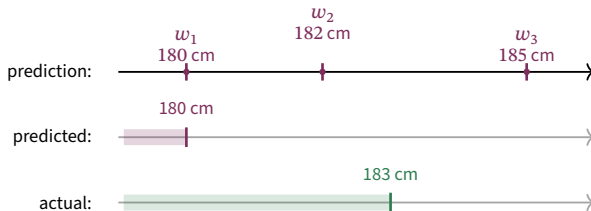
- **Context:** I predicted that Ann's height would be between 180 and 185 cm.
- What the sketched analysis predicts:



Expect in *than*-clauses

(23) Ann is taller than I predicted.

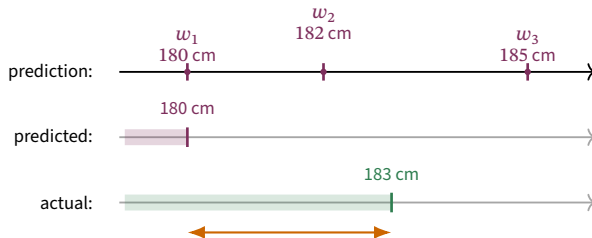
- **Context:** I predicted that Ann's height would be between 180 and 185 cm.
- What the sketched analysis predicts:



Expect in *than*-clauses

(23) Ann is taller than I predicted.

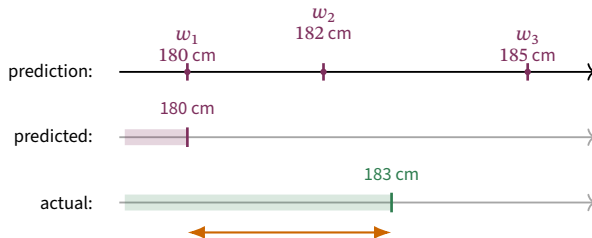
- **Context:** I predicted that Ann's height would be between 180 and 185 cm.
- What the sketched analysis predicts:



Expect in *than*-clauses

(23) Ann is taller than I predicted.

- **Context:** I predicted that Ann's height would be between 180 and 185 cm.
- What the sketched analysis predicts:

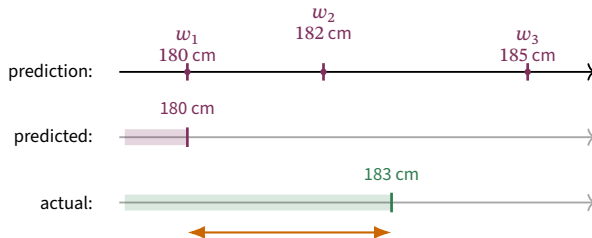


⇒ The analysis compares Ann's actual height with the **minimal** prediction.

Expect in *than*-clauses

(23) Ann is taller than I predicted.

- **Context:** I predicted that Ann's height would be between 180 and 185 cm.
- What the sketched analysis predicts:



⇒ The analysis compares Ann's actual height with the **minimal** prediction.

- Here it wouldn't make sense to say *predict* QRs out of the *than*-clause...

Part 3

A partial solution

The A-not-A analysis

- Consider our entry for *er*:

$$(24) \quad \llbracket \text{er} \rrbracket = \lambda D. \lambda D'. \lambda d. D'(d) \wedge \neg D(d)$$

The A-not-A analysis

- Consider our entry for *er*:

$$(24) \quad \llbracket er \rrbracket = \lambda D. \lambda D'. \lambda d. D'(d) \wedge \neg D(d)$$

- **Schwarzschild's (2008) solution:** syntactically decompose *er*
 - negation is not contributed by *er*
 - negation is instead a scope-taking element within the *than*-clause

The A-not-A analysis

- Consider our entry for *er*:

$$(24) \quad \llbracket er \rrbracket = \lambda D. \lambda D'. \lambda d. D'(d) \wedge \neg D(d)$$

- **Schwarzschild's (2008) solution:** syntactically decompose *er*
 - negation is not contributed by *er*
 - negation is instead a scope-taking element within the *than*-clause

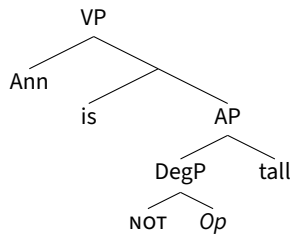
$$(25) \quad \begin{array}{ll} \text{a. } \llbracket er \rrbracket = \lambda D. \lambda D'. \lambda d. D'(d) \wedge D(d) \\ \text{b. } \llbracket \text{NOT} \rrbracket = \lambda d. \lambda D. \neg D(d) \end{array}$$

(see also Gajewski 2008, Alrenga and Kennedy 2014)

Interpreting basic *than*-clauses, I

- We assume that the NOT is base-generated in Spec,AP:

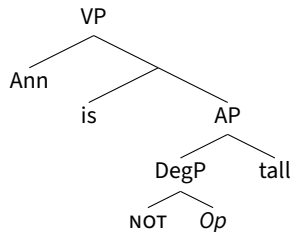
(26)



Interpreting basic *than*-clauses, I

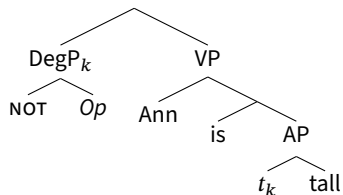
- We assume that the NOT is base-generated in Spec,AP:

(26)



- DegP QRs, carrying NOT along with it:

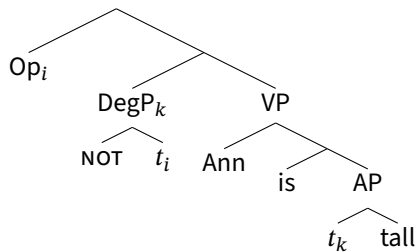
(27)



Interpreting basic *than*-clauses, II

- Finally, Op itself must leave $DegP$:

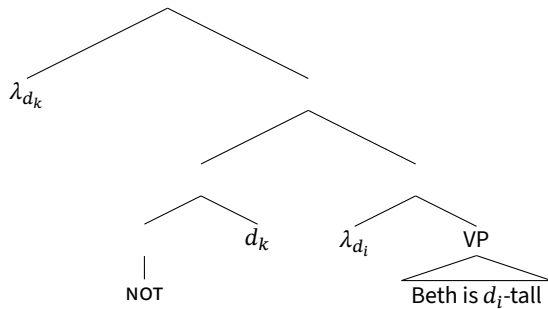
(28)



Interpreting basic *than*-clauses, III

- The structure is interpreted bottom-up:

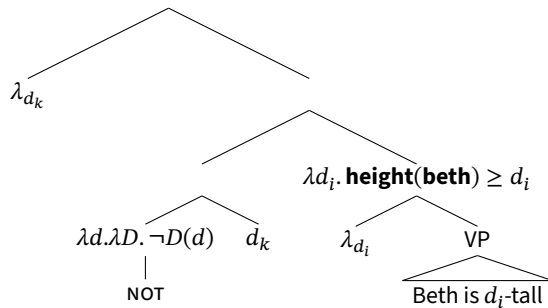
(29)



Interpreting basic *than*-clauses, III

- The structure is interpreted bottom-up:

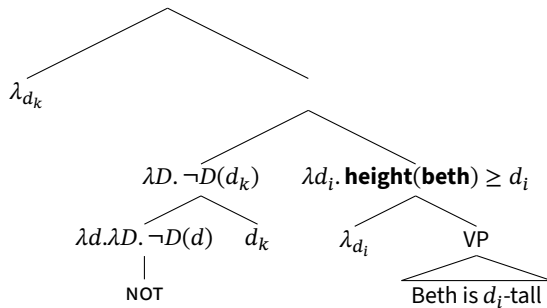
(29)



Interpreting basic *than*-clauses, III

- The structure is interpreted bottom-up:

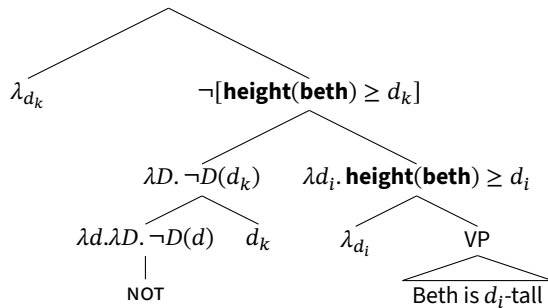
(29)



Interpreting basic *than*-clauses, III

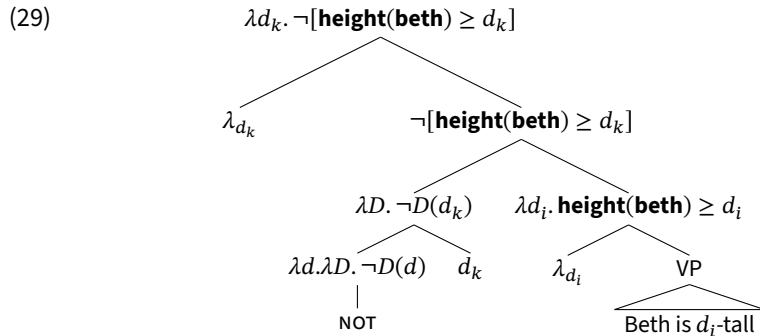
- The structure is interpreted bottom-up:

(29)



Interpreting basic *than*-clauses, III

- The structure is interpreted bottom-up:

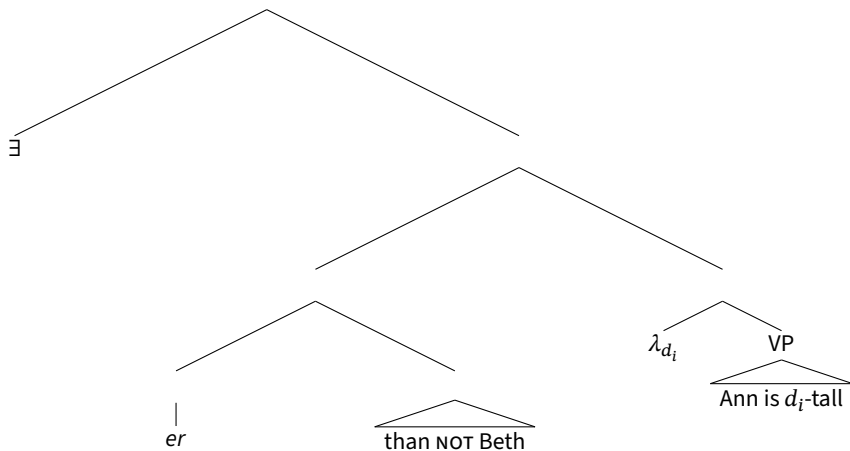


⇒ the *than*-clause denotes the degrees to which Beth is **not** tall.

Interpreting the full sentence

- Interpreting the full sentence:

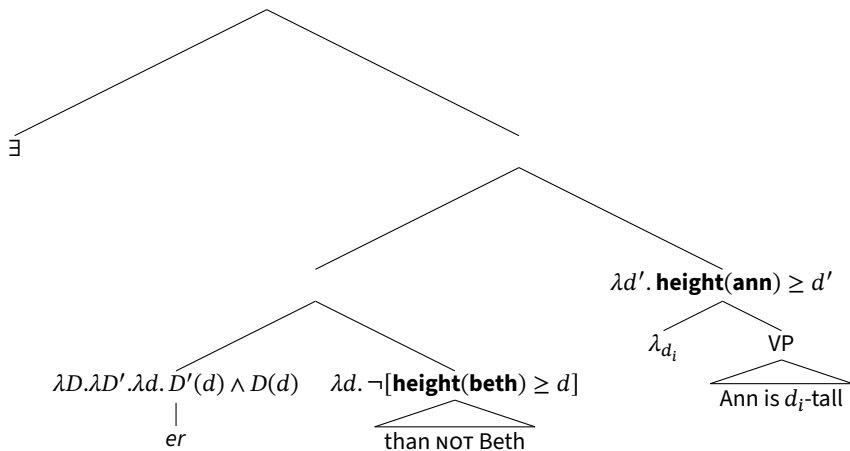
(30)



Interpreting the full sentence

- Interpreting the full sentence:

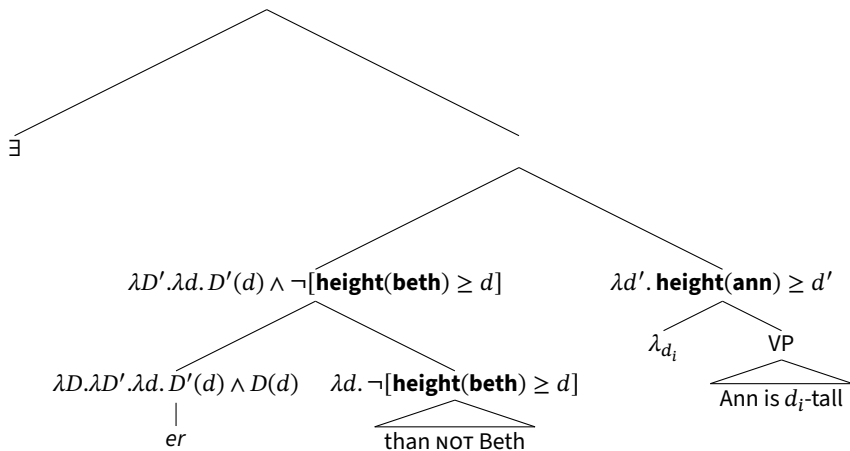
(30)



Interpreting the full sentence

- Interpreting the full sentence:

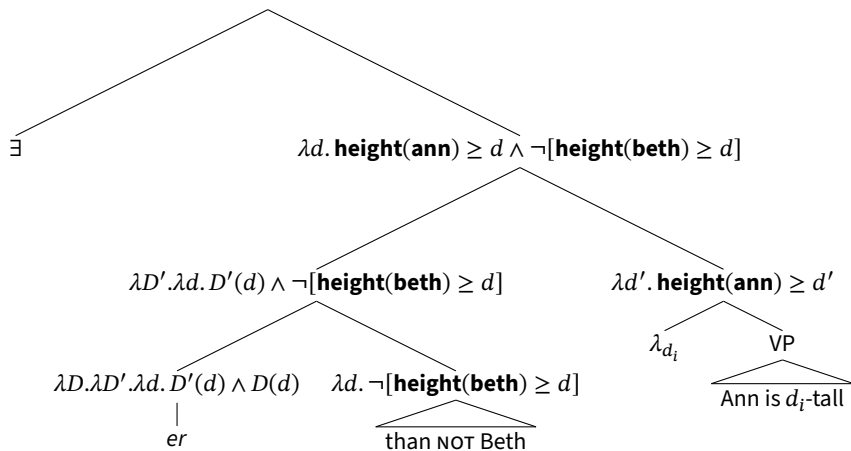
(30)



Interpreting the full sentence

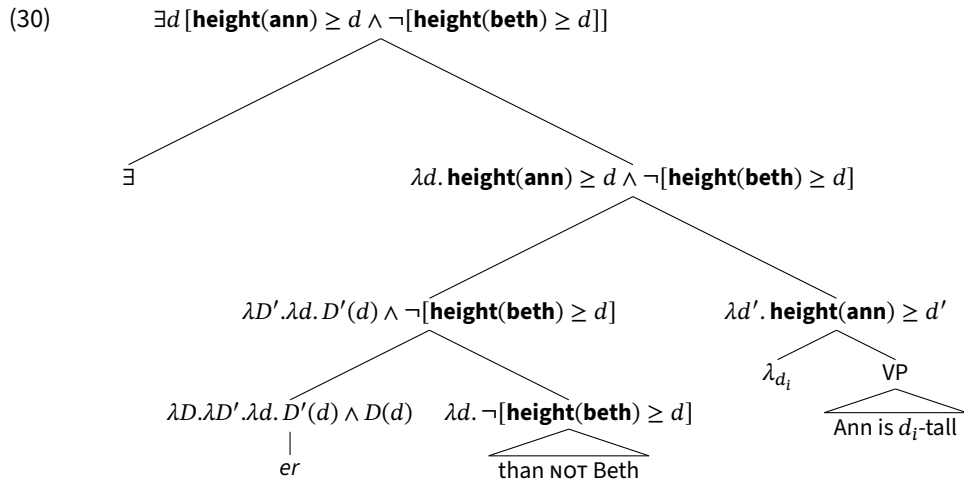
- Interpreting the full sentence:

(30)



Interpreting the full sentence

- Interpreting the full sentence:



Expletive negation in comparatives

- Seuren (1973), who first proposed the A-not-A analysis, observed that negative elements do sometimes show up in *than*-clauses:

(31) French

Jean est plus grand que je **ne** pensais.
Jean be.PRS.3SG more tall than I EXPL.NEG think.IMPF.1SG

‘Jean is taller than I thought.’

(32) Italian

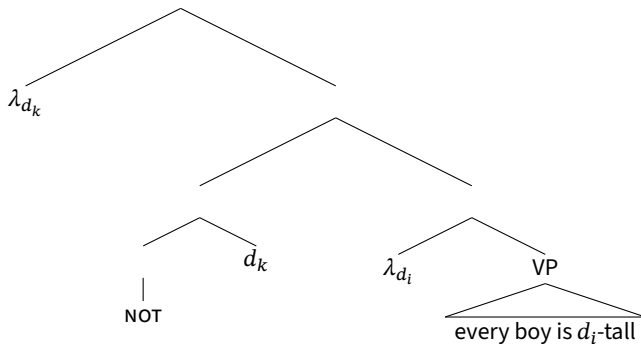
Giovanni è più alto che **non** pensassi.
Giovanni be.PRS.3SG more tall than EXPL.NEG think.IMPF.SBJV.1SG

‘Giovanni is taller than I thought.’

Back to every

- Suppose *every* is interpreted under the scope of NOT:

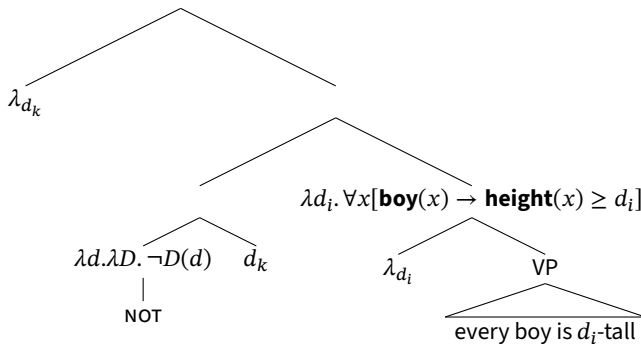
(33)



Back to every

- Suppose *every* is interpreted under the scope of NOT:

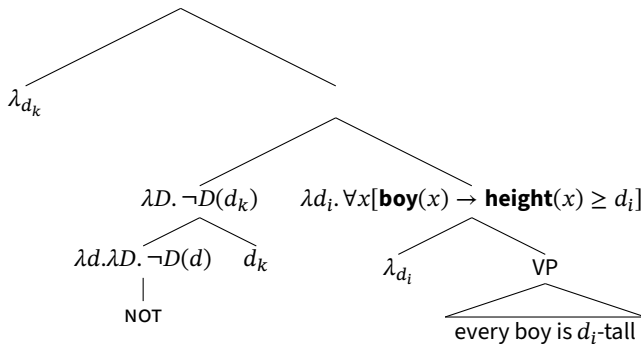
(33)



Back to every

- Suppose *every* is interpreted under the scope of NOT:

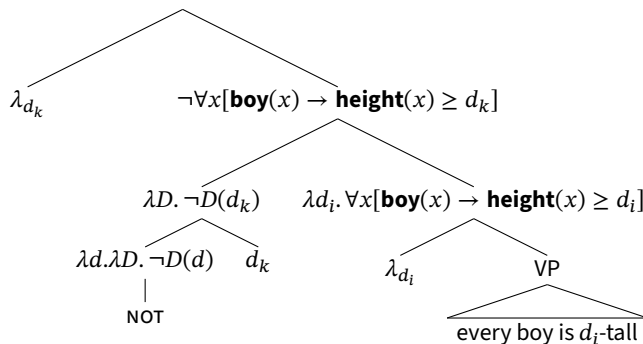
(33)



Back to every

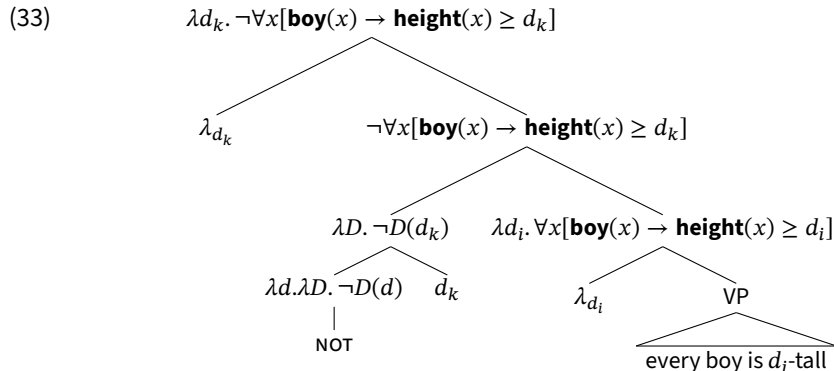
- Suppose *every* is interpreted under the scope of NOT:

(33)



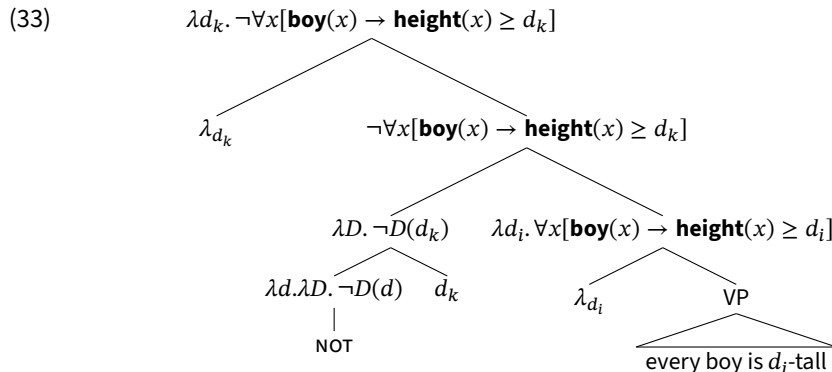
Back to every

- Suppose *every* is interpreted under the scope of NOT:



Back to every

- Suppose *every* is interpreted under the scope of NOT:

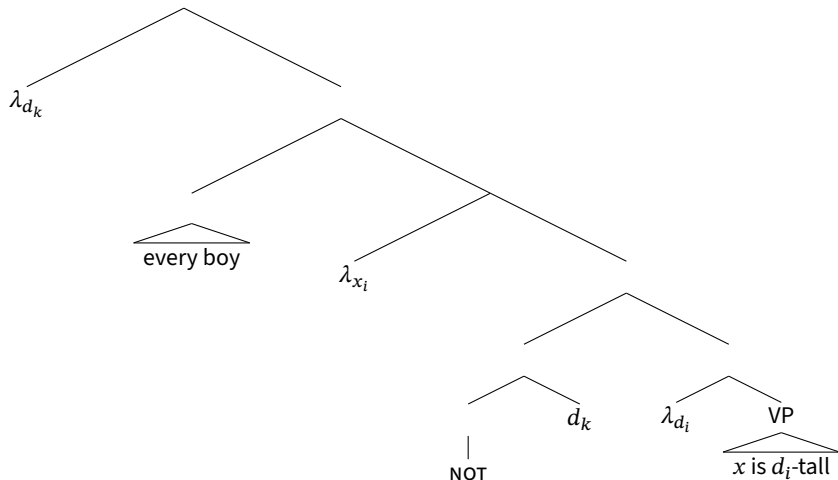


- Which interval is this?

every $\gg$ **NOT**

- But now consider the following structure:

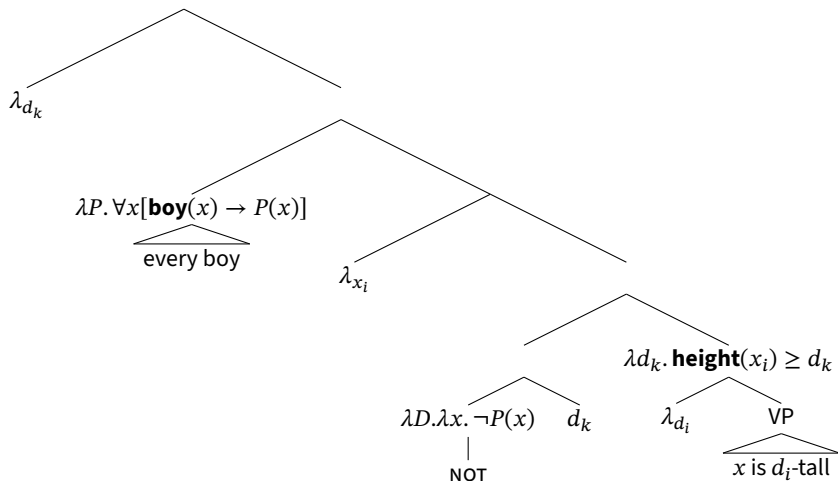
(34)



every $\gg$ **NOT**

- But now consider the following structure:

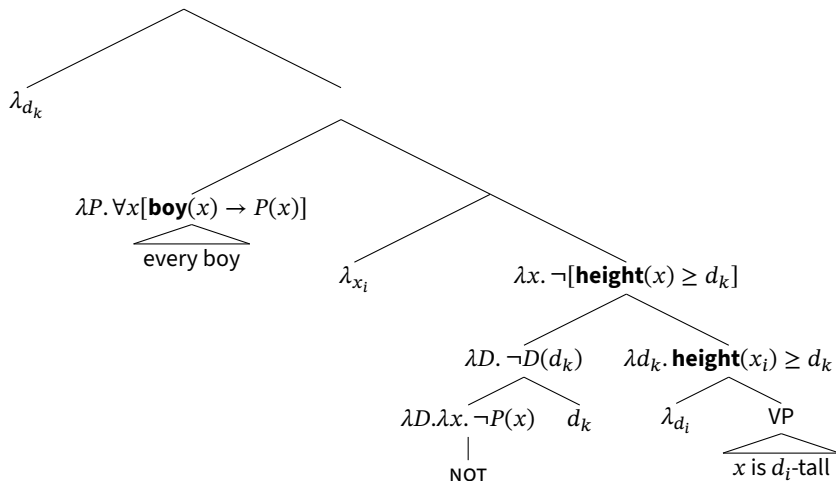
(34)



every $\gg$ **NOT**

- But now consider the following structure:

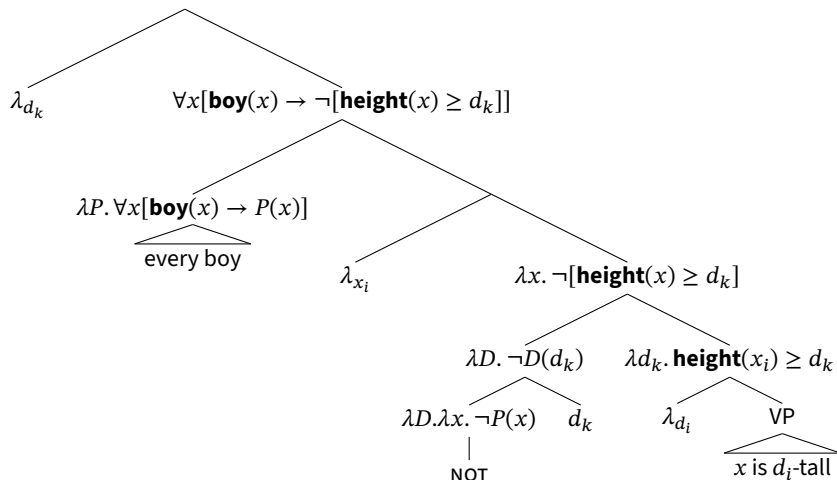
(34)



every $\gg$ **NOT**

- But now consider the following structure:

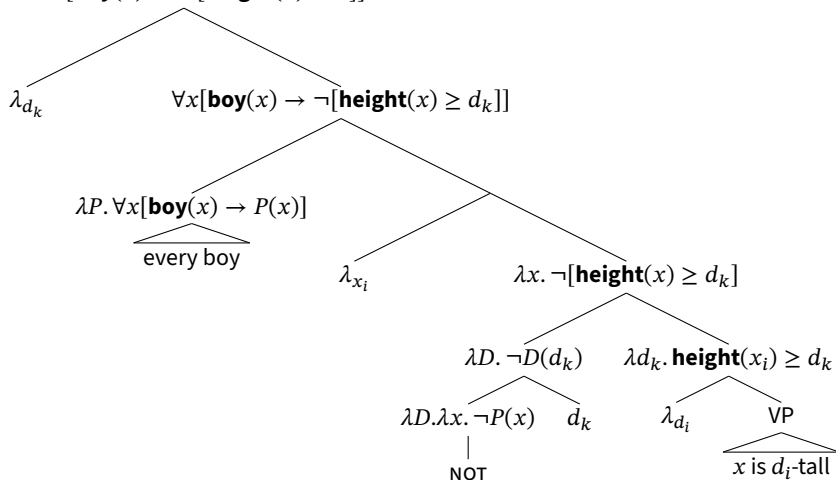
(34)



every $\gg$ **NOT**

- But now consider the following structure:

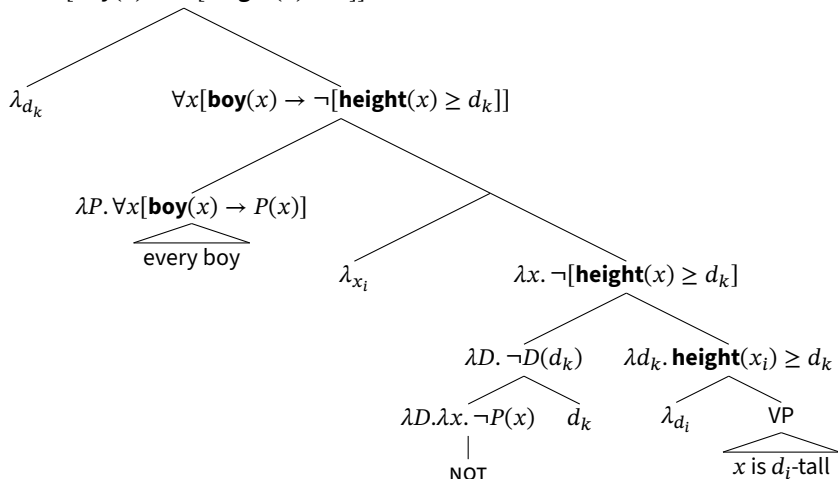
(34) $\lambda d. \forall x [\mathbf{boy}(x) \rightarrow \neg[\mathbf{height}(x) \geq d]]$



every $\gg$ **NOT**

- But now consider the following structure:

(34) $\lambda d. \forall x [\mathbf{boy}(x) \rightarrow \neg[\mathbf{height}(x) \geq d]]$

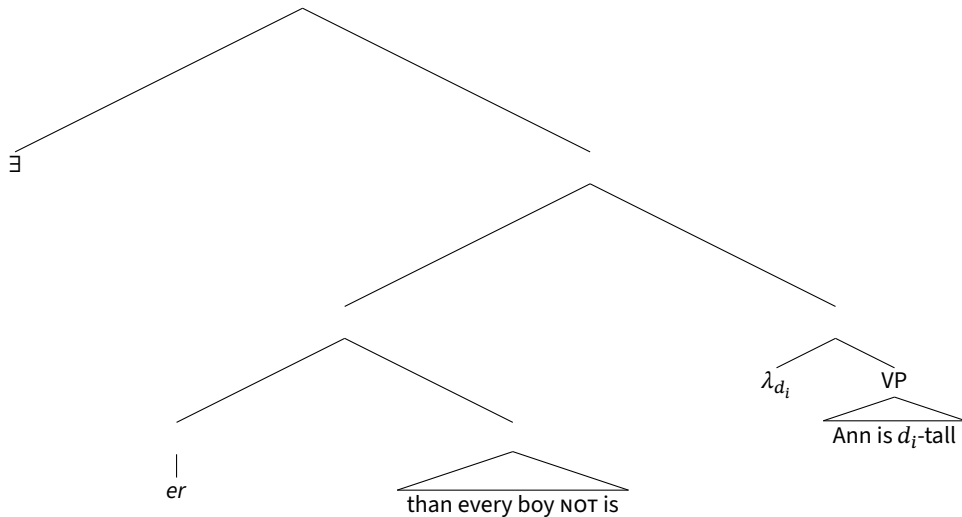


- Which interval is this?

Interpreting the full sentence

- Putting the pieces together:

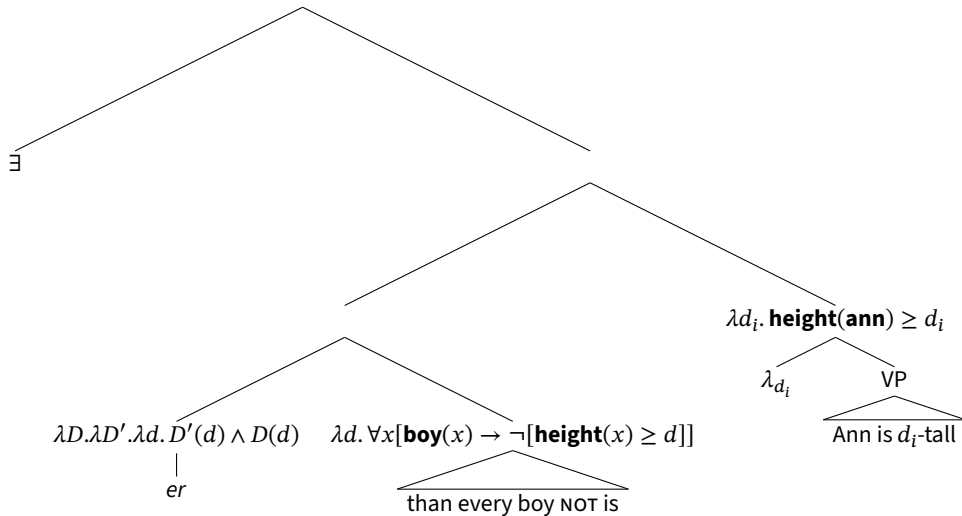
(35)



Interpreting the full sentence

- Putting the pieces together:

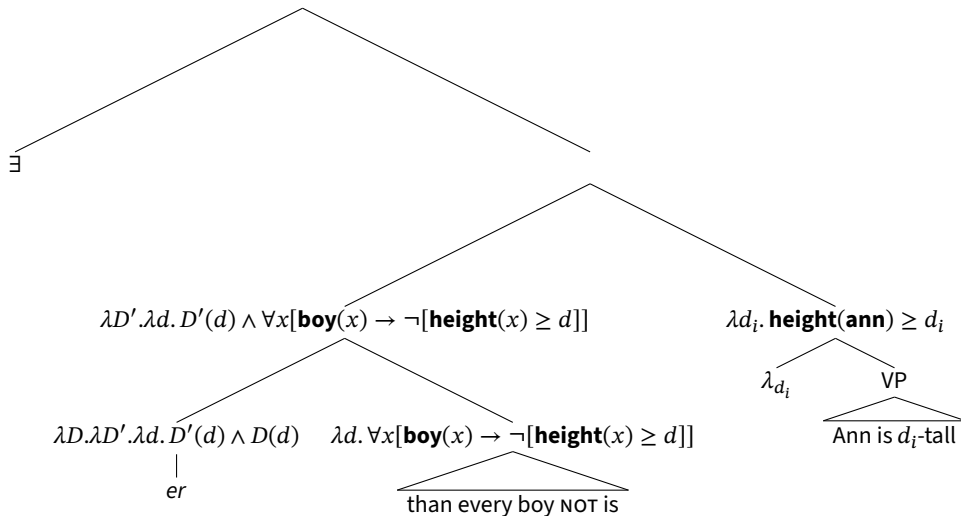
(35)



Interpreting the full sentence

- Putting the pieces together:

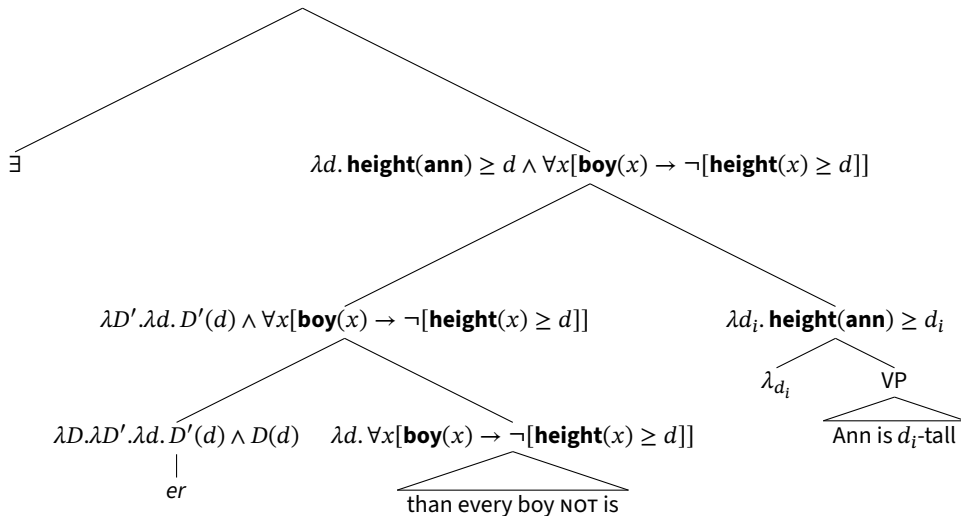
(35)



Interpreting the full sentence

- Putting the pieces together:

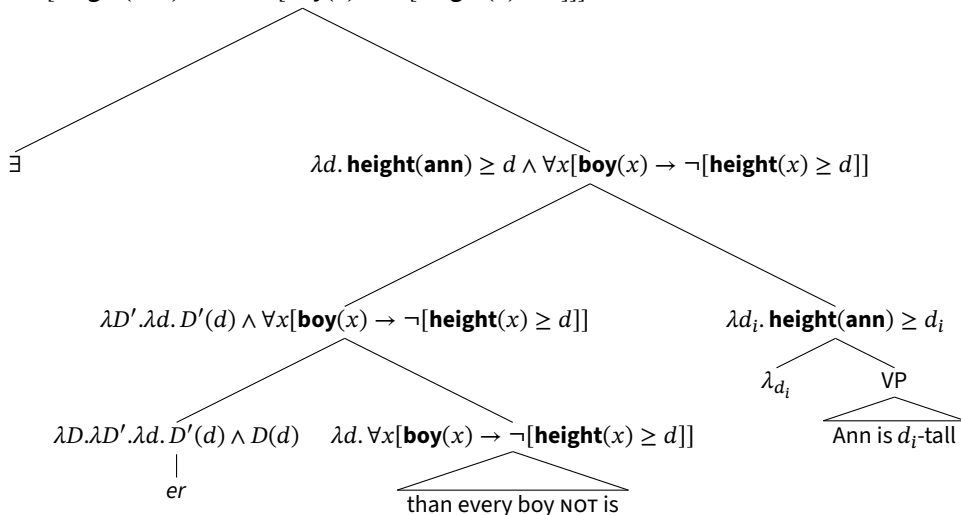
(35)



Interpreting the full sentence

- Putting the pieces together:

(35) $\exists d[\mathbf{height(ann)} \geq d \wedge \forall x[\mathbf{boy}(x) \rightarrow \neg[\mathbf{height}(x) \geq d]]]$



Accounting for *predicted*

- NOT can also scope under *predict*:

(36) **Context:** *I predicted that Ann's height would be between 180 and 185 cm.*

Ann is taller than I predicted.

(37) than λ_d [I predicted [[NOT d] $\lambda_{d'}$ [Ann is d' -tall]]]

Accounting for *predicted*

- NOT can also scope under *predict*:

(36) **Context:** *I predicted that Ann's height would be between 180 and 185 cm.*

Ann is taller than I predicted.

(37) than λ_d [I predicted [[NOT d] $\lambda_{d'}$ [Ann is d' -tall]]]

- The entire sentence would be roughly interpreted as:

(38) $\exists d$ [Ann's height $\geq d$ & I predicted (Ann's height $< d$)]

Part 4

Remaning problems

We can generate the readings but...

- While we can generate the unattested interpretations, we still don't explain why unattested readings are unavailable.

Ex : *why can't 'every' scope under NOT?*

We can generate the readings but...

- While we can generate the unattested interpretations, we still don't explain why unattested readings are unavailable.

Ex : *why can't 'every' scope under NOT?*

- As Schwarzschild 2008 points out, the following sentence is assigned an incorrect reading:

(39) a. Ann is taller than exactly two boys.

b. $\exists d. \mathbf{height(ann)} \geq d \wedge \llbracket \text{exactly two boys} \rrbracket (\lambda x. \mathbf{height}(x) < d)$

We can generate the readings but...

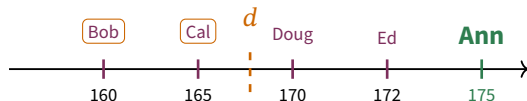
- While we can generate the unattested interpretations, we still don't explain why unattested readings are unavailable.

Ex : *why can't 'every' scope under NOT?*

- As Schwarzschild 2008 points out, the following sentence is assigned an incorrect reading:

(39) a. Ann is taller than exactly two boys.

b. $\exists d. \text{height}(\text{ann}) \geq d \wedge \llbracket \text{exactly two boys} \rrbracket (\lambda x. \text{height}(x) < d)$



exactly two boys below d

– height +

Differentials again

- Once again, we can use differentials to show inadequacies of our proposal:
- (40) Ann is less than 3 years older than every boy.
 $\rightsquigarrow$ *for every boy x , Ann is less than 3 years older than x*

Differentials again

- Once again, we can use differentials to show inadequacies of our proposal:

(40) Ann is less than 3 years older than every boy.

$\rightsquigarrow$ *for every boy x , Ann is less than 3 years older than x*

- We'd interpret the sentence as follows:

(41) **LessThan3yr**($\lambda d. \mathbf{age}(\mathbf{ann}) \geq d \wedge \forall x[\mathbf{boy}(x) \rightarrow \mathbf{age}(x) < d]$)

Differentials again

- Once again, we can use differentials to show inadequacies of our proposal:

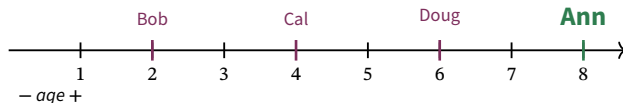
(40) Ann is less than 3 years older than every boy.

$\rightsquigarrow$ for every boy x , Ann is less than 3 years older than x

- We'd interpret the sentence as follows:

(41) **LessThan3yr**($\lambda d. \mathbf{age}(\mathbf{ann}) \geq d \wedge \forall x[\mathbf{boy}(x) \rightarrow \mathbf{age}(x) < d]$)

- To see the issue, consider the following scenario:



Differentials again

- Once again, we can use differentials to show inadequacies of our proposal:

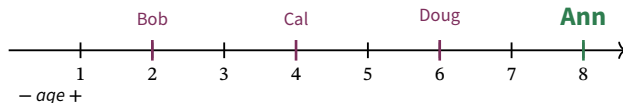
(40) Ann is less than 3 years older than every boy.

$\rightsquigarrow$ for every boy x , Ann is less than 3 years older than x

- We'd interpret the sentence as follows:

(41) **LessThan3yr**($\lambda d. \mathbf{age(ann)} \geq d \wedge \forall x[\mathbf{boy}(x) \rightarrow \mathbf{age}(x) < d]$)

- To see the issue, consider the following scenario:



(42) $\{d \mid \mathbf{age(ann)} \geq d \wedge \forall x[\mathbf{boy}(x) \rightarrow \mathbf{age}(x) < d]\} = \{ \quad \quad \quad \}$

$\Rightarrow$ The analysis measures only the difference between Ann and the **oldest boy**: 2 years.

Differentials again

- Once again, we can use differentials to show inadequacies of our proposal:

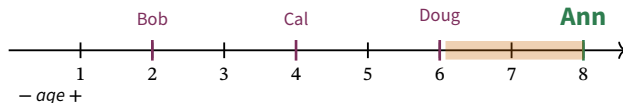
(40) Ann is less than 3 years older than every boy.

$\rightsquigarrow$ for every boy x , Ann is less than 3 years older than x

- We'd interpret the sentence as follows:

(41) **LessThan3yr**($\lambda d. \mathbf{age(ann)} \geq d \wedge \forall x[\mathbf{boy}(x) \rightarrow \mathbf{age}(x) < d]$)

- To see the issue, consider the following scenario:



(42) $\{d \mid \mathbf{age(ann)} \geq d \wedge \forall x[\mathbf{boy}(x) \rightarrow \mathbf{age}(x) < d]\} = \{d_{\mathbf{seven}}, d_{\mathbf{eight}}\}$

$\Rightarrow$ The analysis measures only the difference between Ann and the **oldest boy**: 2 years.

Different approaches

- Heim 2006 provides more empirically adequate accounts of quantifiers within *than*-clauses.
- It also relies on there being two positions quantificational expressions can take scope within a *than*-clause.
- However, it is still unclear why each quantifier behaves the way that they do.

Sentences to think about

- What happens when we combine operators that should take scope at different positions within a *than*-clause?
- (43) Ann is older than any of these girls and any of those ones.
- (44) Ann is older than every one of these boy or every one of those ones.

References I

- Alrenga, Peter and Christopher Kennedy (2014).** “No More Shall We Part: Quantifiers in English Comparatives”. In: *Natural Language Semantics* 22.1, pp. 1–53.
- Gajewski, Jon (2008).** “More on Quantifiers in Comparative Clauses”. In: *Proceedings of Semantics and Linguistic Theory* 18. Ed. by Tova Friedman and Satoshi Ito, pp. 340–357.
- Heim, Irene (Jan. 2006).** “Remarks on Comparative Clauses as Generalized Quantifiers”. Draft manuscript, Massachusetts Institute of Technology.
- Schwarzschild, Roger (2008).** “The Semantics of Comparatives and Other Degree Constructions”. In: *Language and Linguistics Compass* 2.2, pp. 308–331.
- Schwarzschild, Roger and Karina Wilkinson (2002).** “Quantifiers in Comparatives: A Semantics of Degree Based on Intervals”. In: *Natural Language Semantics* 10.1, pp. 1–41.
- Seuren, Pieter A. M. (1973).** “The Comparative”. In: *Generative Grammar in Europe*. Ed. by F. Kiefer and N. Ruwet. Dordrecht: Riedel.